

MTH 203 Lecture 7

Directional Derivative:

We have defined partial derivatives for $f: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$

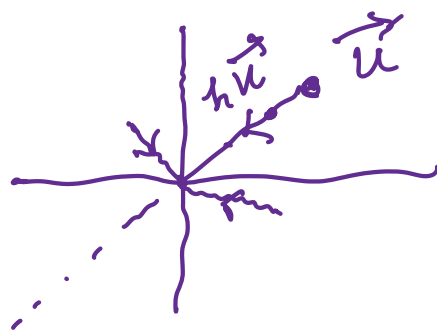
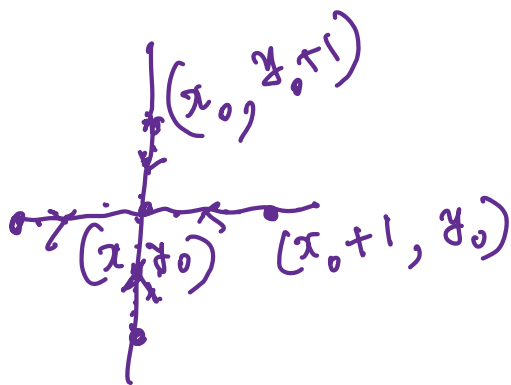
$$\frac{\partial f}{\partial x} \Big|_{(x_0, y_0)} \quad \text{and} \quad \frac{\partial f}{\partial y} \Big|_{(x_0, y_0)}$$

$$\frac{\partial f}{\partial x} \Big|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h} \quad \text{if it exists}$$

$$= \lim_{h \rightarrow 0} \frac{f((x_0, y_0) + h(1, 0)) - f(x_0, y_0)}{h}$$

$$\frac{\partial f}{\partial y} \Big|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h} \quad \text{if it exists}$$

$$= \lim_{h \rightarrow 0} \frac{f((x_0, y_0) + h(0, 1)) - f(x_0, y_0)}{h}$$



Definition: Let $f: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$

and $(x_0, y_0) \in \text{int}(D)$

Let $\vec{u}$ be a unit vector in $\mathbb{R}^2$

We say that f has a "directional derivative" in the direction of $\vec{u}$ at the point (x_0, y_0)

if $\lim_{h \rightarrow 0} \frac{f((x_0, y_0) + h\vec{u}) - f(x_0, y_0)}{h}$ exists

and is denoted by

$$D_{\vec{u}} f \big|_{(x_0, y_0)} \quad \text{or} \quad f'_{\vec{u}}((x_0, y_0))$$

$$\text{or} \quad f'(\vec{u}, (x_0, y_0))$$

Note that $\frac{\partial f}{\partial x} \big|_{(x_0, y_0)} = D_{\vec{e}_1} f \big|_{(x_0, y_0)} = f'_{\vec{e}_1}((x_0, y_0))$

$$\left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)} = D_{\vec{e}_2} f \Big|_{(x_0, y_0)} = f'_{\vec{e}_2}(x_0, y_0)$$

Where $\vec{e}_1 = (1, 0)$, $\vec{e}_2 = (0, 1)$.

Ex: Find the directional derivative of $f(x, y) = 2x^2 + y^2$ at $(-1, 1)$ in the direction of $(3i - 4j)$

Clearly $\vec{u} = \left(\frac{3}{\sqrt{3^2 + 4^2}}, \frac{-4}{\sqrt{3^2 + 4^2}} \right) = \left(\frac{3}{5}, \frac{-4}{5} \right)$

$$\lim_{h \rightarrow 0} \frac{f\left((-1, 1) + h\left(\frac{3}{5}, -\frac{4}{5}\right)\right) - f(-1, 1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f\left(-1 + \frac{3}{5}h, 1 - \frac{4}{5}h\right) - f(-1, 1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2\left(-1 + \frac{3}{5}h\right)^2 + \left(1 - \frac{4}{5}h\right)^2 - [2(-1)^2 + 1^2]}{h}$$

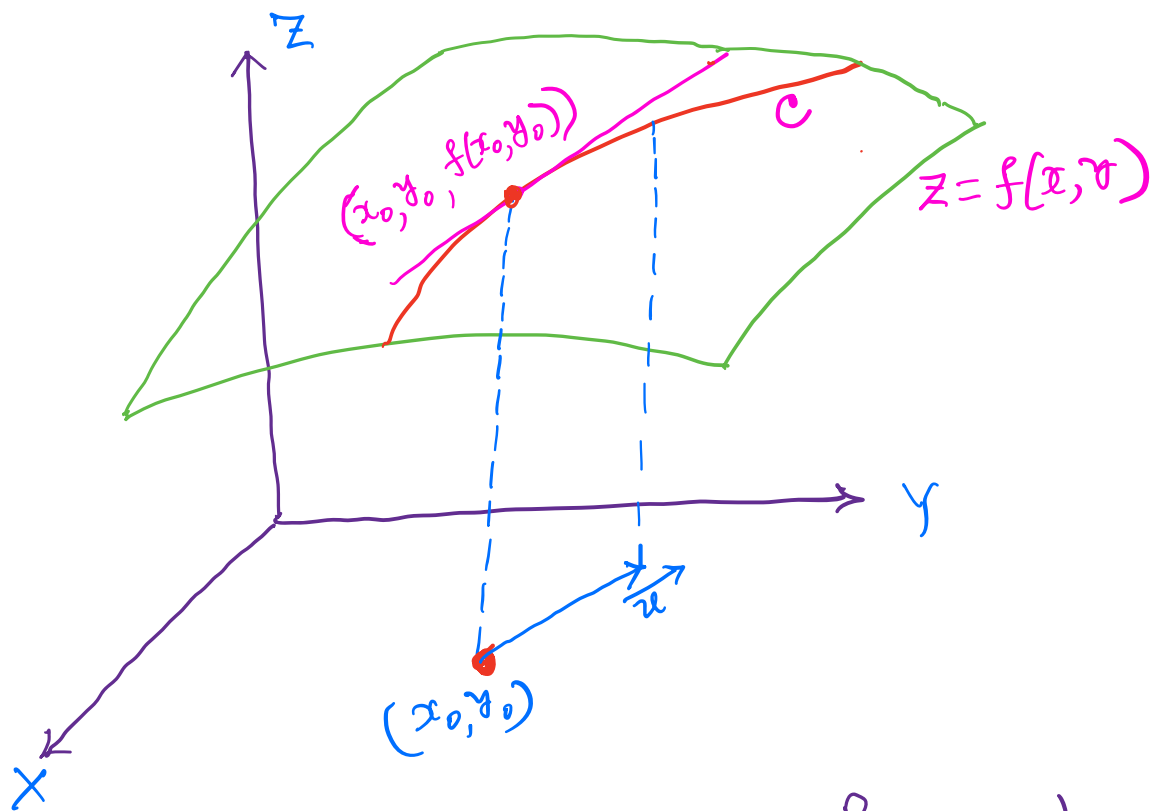
$$= \lim_{h \rightarrow 0} \frac{2\left[1 + \frac{9}{25}h^2 - \frac{6}{5}h\right] + \left(1 + \frac{16}{25}h^2 - \frac{8}{5}h\right) - 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 + \frac{18}{25}h^2 - \frac{12}{5}h + 1 + \frac{16}{25}h^2 - \frac{8}{5}h - 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{34}{25}h^2 - \frac{20}{5}h}{h} = \lim_{h \rightarrow 0} \left(\frac{34}{25}h - 4 \right)$$

$$= \boxed{-4}$$

Geometrical Interpretation of Directional Derivative for $f: \mathbb{R}^2 \rightarrow \mathbb{R}$



- The equation $z = f(x, y)$ represents a surface in 3-dimensional space.
- Let $(x_0, y_0) \in \text{Dom}(f)$ and $(x_0, y_0, f(x_0, y_0))$ be the corresponding point on the surface.
- The vertical plane that passes through (x_0, y_0) and is parallel to $\vec{u}$ intersects the surface in a curve C .
- The rate of change of f in the direction of $\vec{u}$ (at (x_0, y_0)) is the slope of the tangent to C at (x_0, y_0) which is equal to the directional derivative of f in the direction of $\vec{u}$ at (x_0, y_0) .

Ex: Let $f(x, y) = \frac{xy^2}{x^2 + y^4}$ if $x \neq 0$

and $f(0, y) = 0$

Is f continuous at $(0, 0)$?

Along any line ($y = mx$) passing through $(0, 0)$,

$$f(x, y) = \frac{x \cdot m^2 x^2}{x^2 + m^4 x^4} = \frac{x m^2}{1 + m^4 x^2} \text{ goes to zero}$$

as $(x, y) \rightarrow (0, 0)$



Along the parabola $x = y^2$

$$f(x, y) = \frac{y^2 y^2}{y^4 + y^4} = \frac{1}{2} \quad \left(\text{for } (x, y) \neq (0, 0) \right)$$

So, along this parabola $f(x, y) \rightarrow \frac{1}{2}$ as $(x, y) \rightarrow (0, 0)$

Hence $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ doesn't exist and so f is not continuous at $(0, 0)$.



Directional Derivatives:

Let (a_1, a_2) be any unit vector in $\mathbb{R}^2$ and $a_1 \neq 0$

$$\lim_{h \rightarrow 0} \frac{f((0, 0) + h(a_1, a_2)) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{f(a_1 h, a_2 h) - f(0, 0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{a_1 h a_2^2 h^2}{a_1^2 h^2 + a_2^4 h^4} - 0}{h} = \lim_{h \rightarrow 0} \frac{h^3 (a_1 a_2^2)}{h h^2 (a_1^2 + a_2^4 h^2)}$$

$$= \lim_{h \rightarrow 0} \frac{a_1 a_2^2}{a_1^2 + a_2^4 h^2} = \frac{a_1 a_2^2}{a_1^2} = \boxed{\frac{a_2^2}{a_1}} \quad \left(\text{exists since } a_1 \neq 0 \right)$$

Now for $a_1 = 0$,

$$\lim_{h \rightarrow 0} \frac{f((0, 0) + h(0, a_2)) - f(0, 0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(0, h a_2) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = \boxed{0}$$

So, the directional derivative of $f(x, y)$ exists in all directions at $(0, 0)$.

Ex: $f(x, y) = x + y$ if $x = 0$ or $y = 0$
 $= 1$ otherwise

$$\left. \frac{\partial f}{\partial x} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{f(0+h, 0) - f(0, 0)}{h}$$
$$= \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h + 0 - 0}{h} = \boxed{1}$$

$$\left. \frac{\partial f}{\partial y} \right|_{(0,0)} = \lim_{h \rightarrow 0} \frac{f(0, 0+h) - f(0, 0)}{h}$$
$$= \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 + h - 0}{h} = \boxed{1}$$

Is f continuous at $(0, 0)$?

$f(x, y) \rightarrow 0$ as $(x, y) \rightarrow (0, 0)$ along the x or y axis.
 $f(x, y) \rightarrow 1$ as $(x, y) \rightarrow (0, 0)$ along any other line passing through the origin.

So, $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist and

hence f is not continuous at $(0, 0)$.

Directional Derivative

Consider any direction (a_1, a_2) ($a_1 \neq 0, a_2 \neq 0$)

$$\text{Then } \lim_{h \rightarrow 0} \frac{f((0,0) + h(a_1, a_2)) - f(0,0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(a_1 h, a_2 h) - 0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \quad \left(\text{Since } a_1 h \neq 0, a_2 h \neq 0 \right)$$

doesn't exist.

Thus the directional derivative doesn't exist in any direction other than x and y axis.

Total Derivative (Differentiability revisited)

Consider the case of single variable

$$f: D \subseteq \mathbb{R} \longrightarrow \mathbb{R}, \quad x_0 \in \text{int}(D)$$

$\text{int}(D) =$ Set of all interior points of D

f is differentiable at x_0 .

$\Leftrightarrow \exists$ (there exists) a real number (denoted by $f'(x_0)$)

such that $\lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h} = f'(x_0)$

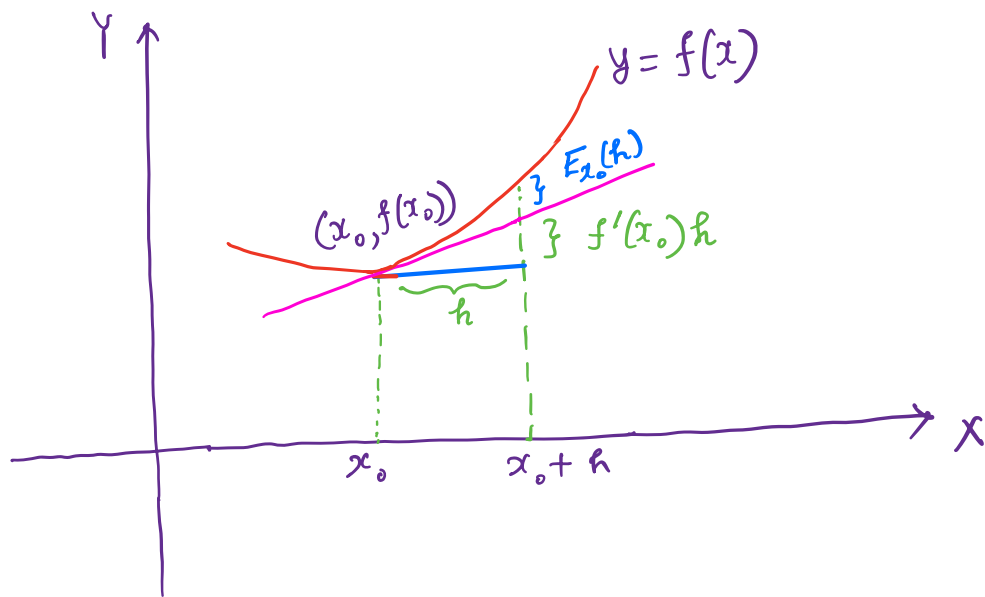
or, $\lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0) - h f'(x_0)}{h} = 0$

Now let us assume that $f'(x_0)$ exists.

Define a function E_{x_0} of h such that

$$E_{x_0}(h) = f(x_0+h) - f(x_0) - f'(x_0)h$$

$$\text{Then } f(x_0+h) - f(x_0) = \underbrace{f'(x_0)h}_{\text{(approximation)}} + \underbrace{E_{x_0}(h)}_{\text{(Error term)}}$$



Thus if f is differentiable at x_0 ,

then
$$\frac{E_{x_0}(h)}{h} \rightarrow 0 \quad \text{as } h \rightarrow 0$$

So, the error $E_{x_0}(h)$ is of smaller order than h .

Note: The derivative $f'(x_0)$ can be thought of as a linear transformation from $\mathbb{R}$ to $\mathbb{R}$ (Multiplication by $f'(x_0)$)

Define $T: \mathbb{R} \rightarrow \mathbb{R}$ by

$$T(h) = f'(x_0)h$$

$$\left(\begin{array}{l} \text{check: } T(h_1 + h_2) = T(h_1) + T(h_2) \\ T(\lambda h) = \lambda T(h) \text{ for } \lambda \in \mathbb{R} \end{array} \right)$$

Another Reformulation

Let $f: D \subseteq \mathbb{R} \rightarrow \mathbb{R}$, $x_0 \in \text{int}(D)$.

f is differentiable at x_0

$\Leftrightarrow \exists$ a linear transformation

$T_{x_0}: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0) - T_{x_0}(h)}{h} = 0$$

(Actually $T_{x_0}(h) = f'(x_0)h$)

When f is differentiable at x_0 ,

$$\text{We set } f'(x_0) = T_{x_0}$$

and call it

“the derivative of f at x_0 ”

This concept allows us to generalize in case of real valued function of several variables.

Definition: Let $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$

$$\text{let } \vec{x}_0 \in \text{int}(D)$$

We say f is differentiable at $\vec{x}_0$
(or f has a total derivative at $\vec{x}_0 \in \mathbb{R}^2$)

$\Leftrightarrow \exists$ a linear transformation

$$T_{\vec{x}_0} : \mathbb{R}^2 \rightarrow \mathbb{R} \text{ such that}$$

$$\lim_{\substack{\vec{h} = (h_1, h_2) \\ \rightarrow (0,0)}} \frac{f(\vec{x}_0 + \vec{h}) - f(\vec{x}_0) - T_{\vec{x}_0}(\vec{h})}{\|\vec{h}\|} = 0$$

$$\text{Note that } \|\vec{h}\| = \sqrt{h_1^2 + h_2^2}$$

When the limit exists,

We set $f'(\vec{x}_0) = T_{x_0}$

and call it

"The total derivative of f at $\vec{x}_0$ "

Note: ① $\lim_{\vec{h} \rightarrow \vec{0}} \frac{E_{\vec{x}_0}(\vec{h})}{\|\vec{h}\|} = 0$

where $E_{\vec{x}_0}(\vec{h}) = f(\vec{x}_0 + \vec{h}) - f(\vec{x}_0) - T_{x_0}(\vec{h})$

So, $E_{\vec{x}_0}(\cdot) : \mathbb{R}^2 \rightarrow \mathbb{R}$

Equivalently we can say:

Let $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$, $\vec{x}_0 \in \text{int}(D)$

f is differentiable at $\vec{x}_0$

if given $\epsilon > 0$, $\exists \delta > 0$ such that

$$\|\vec{h}\| < \delta \Rightarrow \frac{|E_{x_0}(\vec{h})|}{\|\vec{h}\|} < \epsilon$$

or equivalently

$$\|\vec{h}\| < \delta \Rightarrow |E_{x_0}(\vec{h})| < \epsilon \|\vec{h}\|$$

~~~~~ x ~~~~~ x ~~~~~ x ~~~~~ x

Theorem 1: If  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  is differentiable  
at  $\vec{x}_0 \in \mathbb{R}^n$

then there exists linear

transformation  $T_{x_0}: \mathbb{R}^n \rightarrow \mathbb{R}$

$$\text{s.t. } \lim_{\vec{h} \rightarrow \vec{0}} \frac{E_{\vec{x}_0}(\vec{h})}{\|\vec{h}\|} = 0$$

$$\text{where } E_{\vec{x}_0}(\vec{h}) = f(\vec{x}_0 + \vec{h}) - f(\vec{x}_0) - T_{\vec{x}_0}(\vec{h})$$

Theorem 2: Let  $f: \mathbb{R}^2 \rightarrow \mathbb{R}$  be differentiable at  $\vec{x}_0 \in \mathbb{R}^2$  with total derivative  $T_{\vec{x}_0}: \mathbb{R}^2 \rightarrow \mathbb{R}$

then both the partial derivatives of  $f$  at  $\vec{x}_0$  exists and we have

$$T_{\vec{x}_0} = \begin{bmatrix} \frac{\partial f}{\partial x}(\vec{x}_0) & \frac{\partial f}{\partial y}(\vec{x}_0) \end{bmatrix}_{1 \times 2}$$

(Analogous to  $[f'(x_0)]_{1 \times 1}$ )

So,

$$T_{\vec{x}_0}(\vec{h}) = \begin{bmatrix} \frac{\partial f}{\partial x}(\vec{x}_0) & \frac{\partial f}{\partial y}(\vec{x}_0) \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$$

where  $\vec{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$  (Analogous to  $[f'(x_0)][h] = h f'(x_0)$ )

Thus  $T_{\vec{x}_0}(\vec{h}) = h_1 \frac{\partial f}{\partial x}(\vec{x}_0) + h_2 \frac{\partial f}{\partial y}(\vec{x}_0)$

Theorem (3):      Differentiability  
                                  $\Rightarrow$  Continuity.

Let  $f: D \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}$

$\vec{x}_0 \in \text{int}(D)$

If  $f$  is differentiable at  $\vec{x}_0$  then  
 $f$  is continuous at  $\vec{x}_0$